HW 4 due 3/15/24

1) a) $\begin{bmatrix} \pi & -e \\ e & \pi \end{bmatrix}$ $\det(A-\lambda I_2)$ $\lambda I_2 = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$

$\det\begin{bmatrix} \pi-\lambda & -e \\ e & \pi-\lambda \end{bmatrix} = (\pi-\lambda)^2 - (-e\cdot e) = (\pi-\lambda)^2 + e^2 =$

$a=1 \quad b=-2\pi$

$\pi^2 - 2\pi\lambda + \lambda^2 + e^2 = \lambda^2 - 2\pi\lambda + (\pi^2+e^2)$

Discriminant $\Delta = (2\pi)^2 - 4(1)(\pi^2+e^2) = 4\pi^2 - 4\pi^2 - 4e^2 = -4e^2$

Since $\Delta < 0$, there are no real eigenvalues.

$-1+\lambda+\lambda^2-\lambda-1$

(λ^2-2)

b) $\det(A-\lambda I_3)$ $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ $\lambda I_3 = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$ $\det\begin{bmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & -1-\lambda \end{bmatrix}$

$\det(B_{21}) = 0$

$\det(B_{22}) = (1-\lambda)(-1-\lambda)-1$

$\det(B_{23}) = 0$

expand along row 2: $0 + (-1)^{2+2}(1-\lambda)(\lambda^2-1) + 0 = (1-\lambda)(\lambda^2-1) = 0$

$1-\lambda = 0$ $\quad \lambda^2 = 2$

$1 = \lambda$ $\quad \lambda = \pm\sqrt{2}$

$\lambda = -\sqrt{2}, \sqrt{2}, 1$

c) $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ $\lambda I_3 = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$ $\det\begin{bmatrix} 1-\lambda & 1 & 0 \\ 1 & 1-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{bmatrix}$

$\det(B_{11}) = (1-\lambda)^2 - 1$

$\det(B_{12}) = 1-\lambda$

$\det(B_{13}) = 1$

expand along row 1: $(-1)^{1+1}(1-\lambda)\left[(1-\lambda)^2-1\right] + (-1)^{1+2}(1)(1-\lambda) + (-1)^{1+3}(0)(1) =$

$(1-\lambda)^3 - 1 + \lambda + 1 + \lambda = (1-\lambda)^3 - 2 + 2\lambda = (1-\lambda)^3 - 2(1-\lambda) = 0$

$(1-\lambda)\left[(1-\lambda)^2 - 2\right] = 0$ $\quad (1-\lambda)^2 - 2 = 0$

$1-\lambda = 0$ $\quad (1-\lambda)^2 = 2$

$\lambda = 1$ $\quad 1-\lambda = \pm\sqrt{2}$

$\lambda = 1-\sqrt{2}$ $\quad \lambda = 1+\sqrt{2}$

eigenvalues & eigenvectors:

$\lambda_1 = \sqrt{e\pi}$

$\lambda_2 = -\sqrt{e\pi}$

2) a) $\begin{bmatrix} 0 & \pi \\ e & 0 \end{bmatrix}$ $\lambda I_2 = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$ $\det(A - \lambda I_2) = \begin{bmatrix} -\lambda & \pi \\ e & -\lambda \end{bmatrix} = \lambda^2 - e\pi$

$\lambda^2 - e\pi = 0 \quad \lambda^2 = e\pi \quad \lambda = \pm\sqrt{e\pi}$

By corollary 24.1, A is diagonalizable since there are 2 distinct real roots.

Solve $(A - \lambda_1 I_2)\vec{x} = \vec{0}$, where $\lambda_1 = \sqrt{e\pi}$

$$\left[\begin{array}{cc|c} -\sqrt{e\pi} & \pi & 0 \\ e & -\sqrt{e\pi} & 0 \end{array}\right] \xrightarrow{r_1\left(\frac{-1}{\sqrt{e\pi}}\right)} \left[\begin{array}{cc|c} 1 & \frac{-\pi}{\sqrt{e\pi}} & 0 \\ e & -\sqrt{e\pi} & 0 \end{array}\right] \xrightarrow{r_2 - r_1(e)} \left[\begin{array}{cc|c} 1 & \frac{-\pi}{\sqrt{e\pi}} & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$\begin{cases} x_1 = \frac{\pi}{\sqrt{e\pi}} x_2 \\ x_2 \text{ is free} \end{cases}$ $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \pi/\sqrt{e\pi} \cdot x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} \frac{\pi}{\sqrt{e\pi}} \\ 1 \end{bmatrix}$ $E_{\sqrt{e\pi}}$

$\lambda_2 = -\sqrt{e\pi}$

$$\left[\begin{array}{cc|c} \sqrt{e\pi} & \pi & 0 \\ e & \sqrt{e\pi} & 0 \end{array}\right] \xrightarrow{r_1\left(\frac{1}{\sqrt{e\pi}}\right)} \left[\begin{array}{cc|c} 1 & \frac{\pi}{\sqrt{e\pi}} & 0 \\ e & \sqrt{e\pi} & 0 \end{array}\right] \xrightarrow{r_2 - r_1(e)} \left[\begin{array}{cc|c} 1 & \frac{\pi}{\sqrt{e\pi}} & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$\begin{cases} x_1 = \frac{-\pi}{\sqrt{e\pi}} x_2 \\ x_2 \text{ is free} \end{cases}$ $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\pi/\sqrt{e\pi} \cdot x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} \frac{-\pi}{\sqrt{e\pi}} \\ 1 \end{bmatrix}$ $E_{-\sqrt{e\pi}}$

$E_{\sqrt{e\pi}} = \left\{ \begin{bmatrix} \frac{\pi}{\sqrt{e\pi}} \\ 1 \end{bmatrix} \right\}$ $E_{-\sqrt{e\pi}} = \left\{ \begin{bmatrix} -\pi/\sqrt{e\pi} \\ 1 \end{bmatrix} \right\}$

$S = \begin{bmatrix} \pi/\sqrt{e\pi} & -\pi/\sqrt{e\pi} \\ 1 & 1 \end{bmatrix}$ $D = \begin{bmatrix} \sqrt{e\pi} & 0 \\ 0 & -\sqrt{e\pi} \end{bmatrix}$

b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ $\lambda I_3 = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$ $\det(A - \lambda I_3) = \det \begin{bmatrix} 1-\lambda & 1 & 1 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{bmatrix} = (1-\lambda)^3$

almu(1) = 3 $(1-\lambda)^3 = 0$ $(A - \lambda_1 I_3)\vec{x} = 0$

$1 - \lambda = 0$

$\lambda = 1$

$$\left[\begin{array}{ccc|c} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\begin{cases} x_1 \text{ is free} \\ x_2 = 0 \\ x_3 = 0 \end{cases}$

$x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $E_1 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$ gemu(1) = 1

gemu = almu = diagonalizable

Since gemu(1) = 1 < 3 = almu(1), the matrix is not diagonalizable

$ab > 0$

$a = b = 0$

4) $\begin{bmatrix} 1 & a \\ b & 1 \end{bmatrix}$ $\det(A - \lambda I_2) = \det \begin{bmatrix} 1-\lambda & a \\ b & 1-\lambda \end{bmatrix} = (1-\lambda)^2 - ab$

$(1-\lambda)^2 - ab = 0$

$\lambda^2 - 2\lambda + (1-ab) = 0$

Δ: $(-2)^2 - 4(1)(1-ab) =$

$4 - 4(1-ab) = 4ab$

[cases: diagonalizable
1 $\Delta > 0$: 2 distinct real roots - either a & b both + " " −
2 $\Delta < 0$: only a or only b (-) → no dist roots
3 $\Delta = 0$: 1 real root, w/ almu(2) = 2

• case 3: $\Delta = 0$ $4ab = 0$, so either $a=0, b \neq 0$; $b=0, a \neq 0$; both $ab = 0$

$\lambda^2 - 2\lambda + 1 - (0) = 0$

$\lambda^2 - 2\lambda + 1 = 0$

$(\lambda-1)^2 = 0$

$\boxed{\lambda = 1}$ almu(1) = 2

× • case 3.1 - $a=0$, $b \neq 0$

$\begin{bmatrix} 0 & 0 & 0 \\ b & 0 & 0 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $bx_1 = 0$, x_2 free

$E_1 = \left\{ s \begin{pmatrix} 0 \\ 1 \end{pmatrix} : s \in \mathbb{R} \right\}$

gemu(1) = 1 < 2 = almu(1) so not diagonalizable

× • case 3.2 $a \neq 0$, $b = 0$ $\begin{bmatrix} 0 & a & 0 \\ 0 & 0 & 0 \end{bmatrix}$ x_1 free, $x_2 = 0$

$E_1 = \left\{ s \begin{pmatrix} 1 \\ 0 \end{pmatrix} : s \in \mathbb{R} \right\}$ gemu(1) = 1 < 2 = almu(1)

so not diagonalizable

how to find gemu ✓

• case 3.3 $a = 0 = b$ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ x_1 free, x_2 free

$\ker \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \mathbb{R}^2$ $\dim(\mathbb{R}^2) = 2$

almu(1) = 2 = gemu(1) so diagonalizable

$ab > 0$

A is diagonalizable when either ① a and b have the same sign (nonzero)
② $a = 0 = b$

3) a) $\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$ $\det(A - \lambda I_2) = \det \begin{bmatrix} \frac{1}{2}-\lambda & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2}-\lambda \end{bmatrix} = (\frac{1}{2}-\lambda)^2 - \frac{1}{4} = 0 \Rightarrow (\frac{1}{2}-\lambda)^2 = \frac{1}{4}$

$\frac{1}{2} - \lambda = \pm \frac{1}{2}$

$\frac{1}{2} - \lambda = \frac{1}{2}$ $\lambda = 0$; $\frac{1}{2} - \lambda = -\frac{1}{2}$ $\lambda = 1$

$\lambda_1 = 0$ $(A - \lambda_1 I_2)\vec{x} = 0$

$\begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \end{bmatrix} \xrightarrow{r_2 - r_1} \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\frac{1}{2}x_1 + \frac{1}{2}x_2 = 0$

$x_1 = -x_2$, x_2 is free $\quad x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $E_0 = \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$

$\lambda_2 = 1$

$\begin{bmatrix} -1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \end{bmatrix} \xrightarrow{r_2 + r_1} \begin{bmatrix} -1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $-\frac{1}{2}x_1 + \frac{1}{2}x_2 = 0$

$x_1 = x_2$, x_1 free $\quad x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $E_1 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

orthogonal but not unit vectors

$\vec{v_1} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $|\vec{v_1}| = \sqrt{2}$

$\vec{v_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\vec{u_1} = \frac{\vec{v_1}}{|\vec{v_1}|} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

proj: $\vec{v_2} - (\vec{v_2} \cdot \vec{u_1})\vec{u_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - (0)\vec{u_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\vec{u_2} = \frac{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\sqrt{2}} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

$D = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ $S = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

3) b) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ $\det(A-\lambda I_3) = \begin{bmatrix} 1-\lambda & 1 & 0 \\ 1 & 1-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{bmatrix}$ expand along row 1:

$\det a_{11} = (1-\lambda)^2 - 1$

$\det a_{12} = (1-\lambda)\cdot 0 = 1-\lambda$

$\det a_{13} = 1$

$(-1)^{1+1}((1-\lambda)^2-1)(1-\lambda) + (-1)^{1+2}(1-\lambda)(1) + (-1)^{1+3}(1)(0) =$

$[(1-\lambda)^2-1](1-\lambda) + -1 + \lambda = (1-\lambda)^3 - 1 + \lambda - 1 + \lambda = (1-\lambda)^3 + 2\lambda - 2 =$

$(1-\lambda)^3 + 2(\lambda-1) = 0$

$(1-\lambda)(1-2\lambda+\lambda^2) + 2(\lambda-1) = 0$

$(1-\lambda)[(1-2\lambda+\lambda^2) - 2] = 0$

$(1-\lambda)(\lambda^2-2\lambda-1) = 0$

$\lambda = 1$ ← $\frac{2 \pm \sqrt{4-4(1)(-1)}}{2} = \frac{2 \pm \sqrt{8}}{2} = \frac{2 \pm 2\sqrt{2}}{2} = \frac{2(1 \pm \sqrt{2})}{2} = 1+\sqrt{2}$ or $1-\sqrt{2}$

Eigenvals: $\lambda_1 = 1$, $\lambda_2 = 1+\sqrt{2}$, $\lambda_3 = 1-\sqrt{2}$

λ_1: $(A - I_3)\vec{x} = 0$

$\left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{r_3 - r_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$x_1 = -x_3$

$x_1 + x_3 = 0$

$x_2 = 0$

x_3 is free

$x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ $E_1 = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

λ_2: $\left[\begin{array}{ccc|c} -\sqrt{2} & 1 & 0 & 0 \\ 1 & -\sqrt{2} & 1 & 0 \\ 0 & 1 & -\sqrt{2} & 0 \end{array}\right] \xrightarrow{r_1\left(\frac{-1}{\sqrt{2}}\right)} \left[\begin{array}{ccc|c} 1 & \frac{-1}{\sqrt{2}} & 0 & 0 \\ 1 & -\sqrt{2} & 1 & 0 \\ 0 & 1 & -\sqrt{2} & 0 \end{array}\right] \xrightarrow{r_2 - r_1} \left[\begin{array}{ccc|c} 1 & \frac{-1}{\sqrt{2}} & 0 & 0 \\ 0 & -\sqrt{2}+\frac{1}{\sqrt{2}} & 1 & 0 \\ 0 & 1 & -\sqrt{2} & 0 \end{array}\right] \xrightarrow{r_1 + \left(\frac{-1}{\sqrt{2}}\right)r_2}$

$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & \frac{-1}{\sqrt{2}} & 1 & 0 \\ 0 & 1 & -\sqrt{2} & 0 \end{array}\right] \xrightarrow{r_3 + r_2(\sqrt{2})} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & \frac{-1}{\sqrt{2}} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{r_2(-\sqrt{2})} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -\sqrt{2} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$

$x_1 - x_3 = 0$

$x_2 - \sqrt{2}x_3 = 0$

x_3 free

$x_1 = x_3$

$x_2 = \sqrt{2}x_3$

x_3 free

$x_3 \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}$ $E_{1+\sqrt{2}} = \left\{ \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix} \right\}$

λ_3:

$$\left[\begin{array}{ccc|c} \sqrt{2} & 1 & 0 & 0 \\ 1 & \sqrt{2} & 1 & 0 \\ 0 & 1 & \sqrt{2} & 0 \end{array}\right] \xrightarrow{r_1\left(\frac{1}{\sqrt{2}}\right)} \left[\begin{array}{ccc|c} 1 & \frac{1}{\sqrt{2}} & 0 & 0 \\ 1 & \sqrt{2} & 1 & 0 \\ 0 & 1 & \sqrt{2} & 0 \end{array}\right] \xrightarrow{r_2-r_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 1 & 0 \\ 0 & 1 & \sqrt{2} & 0 \end{array}\right] \xrightarrow{r_1-r_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 1 & 0 \\ 0 & 1 & \sqrt{2} & 0 \end{array}\right] \xrightarrow{r_2(\sqrt{2})} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & \sqrt{2} & 0 \\ 0 & 1 & \sqrt{2} & 0 \end{array}\right] \xrightarrow{r_3-r_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & \sqrt{2} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_1 - x_3 = 0 \quad x_1 = x_3$

$x_2 + \sqrt{2}x_3 = 0 \quad x_2 = -\sqrt{2}x_3$

x_3 free $\quad x_3$ free $\quad x_3 \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix}$ $\quad E_{1,-\sqrt{2}} = \left\{ \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix} \right\}$

$\vec{v_1} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = E_1 \quad \vec{u_1} = \frac{\vec{v_1}}{\|\vec{v_1}\|} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$

$\vec{v_2} = \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix} = E_{1,\sqrt{2}} \quad \text{perp}_{\vec{v_1}}\vec{v_2} = \vec{v_2} - (\vec{v_2}\cdot\vec{v_1})\vec{u_1} = \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix} - 0 = \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}$

$\vec{v_3} = \begin{bmatrix} -1 \\ -\sqrt{2} \\ 1 \end{bmatrix} = E_{1,-\sqrt{2}} \quad \vec{u_2} = \frac{\begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}}{2} = \begin{bmatrix} 1/2 \\ \sqrt{2}/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/\sqrt{2} \\ 1/2 \end{bmatrix}$

$\text{perp}_{\vec{v_2}}\vec{v_3} = \vec{v_3} - (\vec{v_3}\cdot\vec{u_1})\vec{u_2} = \begin{bmatrix} -1 \\ -\sqrt{2} \\ 1 \end{bmatrix} - 0 = \begin{bmatrix} -1 \\ -\sqrt{2} \\ 1 \end{bmatrix}$

$\vec{u_3} = \begin{bmatrix} -1/2 \\ -1/\sqrt{2} \\ 1/2 \end{bmatrix} \quad S = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1+\sqrt{2} & 0 \\ 0 & 0 & 1-\sqrt{2} \end{bmatrix}$

6) $A^{2024} \quad A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \quad \det(A - \lambda I_2) = \begin{bmatrix} 2-\lambda & 3 \\ 4 & 1-\lambda \end{bmatrix} = (2-\lambda)(1-\lambda) - 12 = 2 - 3\lambda + \lambda^2 - 12 = \lambda^2 - 3\lambda - 10$

$\lambda^2 - 3\lambda - 10 = (\lambda - 5)(\lambda + 2) = 0 \quad \lambda_1 = 5 \quad \lambda_2 = -2$

$\lambda_1 = 5$:

$$\left[\begin{array}{cc|c} -3 & 3 & 0 \\ 4 & -4 & 0 \end{array}\right] \xrightarrow{r_1\left(\frac{-1}{3}\right)} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 4 & -4 & 0 \end{array}\right] \xrightarrow{r_2 - r_1(4)} \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$x_1 - x_2 = 0 \quad x_1 = x_2 \quad x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

x_2 free

$E_5 = \left\{ s \begin{bmatrix} 1 \\ 1 \end{bmatrix} : s \in \mathbb{R} \right\}$

$\lambda_2 = -2$

$$\left[\begin{array}{cc|c} 4 & 3 & 0 \\ 4 & 3 & 0 \end{array}\right] \xrightarrow{r_1\left(\frac{1}{4}\right)} \left[\begin{array}{cc|c} 1 & 3/4 & 0 \\ 4 & 3 & 0 \end{array}\right] \xrightarrow{r_2 - r_1(4)} \left[\begin{array}{cc|c} 1 & 3/4 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$x_1 + 3/4\, x_2 = 0 \quad x_1 = -\frac{3}{4}x_2 \quad x_2 \begin{bmatrix} -3/4 \\ 1 \end{bmatrix}$

x_2 free

$E_2 = \left\{ s \begin{bmatrix} -3/4 \\ 1 \end{bmatrix} : s \in \mathbb{R} \right\}$

choose x_2 to be $4 \rightarrow x_2 \begin{bmatrix} -3 \\ 4 \end{bmatrix}$

compute $S^{-1} = \frac{1}{4+3}\begin{bmatrix} 4 & 3 \\ -1 & 1 \end{bmatrix} = \frac{-1}{7}\begin{bmatrix} 4 & 3 \\ -1 & 1 \end{bmatrix}$

$S = \begin{bmatrix} 1 & -3 \\ 1 & 4 \end{bmatrix} \quad D^{2024} = \begin{bmatrix} 5^{2024} & 0 \\ 0 & 2^{2024} \end{bmatrix}$

OH 3/13

A is symmetric w/ eigenvalues $\lambda_1 \dots \lambda_n$

find eigenvectors

want $v_1 \dots v_n$ basis to be orthonormal

$D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$

↳ IF v_i, v_j have diff eigenvalues, automatically $v_i \perp v_j$

↳ SO we just do GSP on each eigenspace

6) cont. $S = \begin{bmatrix} 1 & -3 \\ 1 & 4 \end{bmatrix}$ $D^{2024} = \begin{bmatrix} 5^{2024} & 0 \\ 0 & (-2)^{2024} \end{bmatrix}$ $S^{-1} = \begin{bmatrix} 4/7 & 3/7 \\ -1/7 & 1/7 \end{bmatrix}$

$SD^{2024} = \begin{bmatrix} 1 & -3 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 5^{2024} & 0 \\ 0 & -2^{2024} \end{bmatrix} = \begin{bmatrix} 5^{2024} & -3(-2)^{2024} \\ 5^{2024} & 4(-2)^{2024} \end{bmatrix}$

$A^{2024} = SD^{2024}S^{-1} = \begin{bmatrix} 5^{2024} & (-3)(-2)^{2024} \\ 5^{2024} & 4(-2)^{2024} \end{bmatrix}\begin{bmatrix} 4/7 & 3/7 \\ -1/7 & 1/7 \end{bmatrix} =$

$\begin{bmatrix} \frac{4}{7}(5)^{2024} + \frac{3}{7}(-2)^{2024} & \frac{3}{7}5^{2024} - \frac{3}{7}(-2)^{2024} \\ \frac{4}{7}(5)^{2024} - \frac{4}{7}(-2)^{2024} & \frac{3}{7}5^{2024} + (\frac{4}{7})(-2)^{2024} \end{bmatrix}$

ensure $(-2)^{2024} = +$ ↘ since even power

$\begin{bmatrix} (\frac{4}{7})5^{2024} + (\frac{3}{7})2^{2024} & (\frac{3}{7})5^{2024} - (\frac{3}{7})2^{2024} \\ (\frac{4}{7})5^{2024} - (\frac{4}{7})2^{2024} & (\frac{3}{7})5^{2024} + (\frac{4}{7})2^{2024} \end{bmatrix}$ ← final answer

5) $\vec{v_1}, \vec{v_2}, \vec{v_3}, \vec{v_4}, \vec{v_5}$ orthonormal basis of $\mathbb{R}^5$

$V = \text{span}(\vec{v_1}, \vec{v_2}, \vec{v_3})$

P = projection onto V

$P = AA^T$

orthonormal

Basis for $E_{\lambda_1} = V$:

$\{\vec{v_1}, \vec{v_2}, \vec{v_3}\}$

Given $V = \text{span}(\vec{v_1}, \vec{v_2}, \vec{v_3})$

★ orthonormal basis of $E_{\lambda_2} = V^\perp$ λ_2: Eigenspace of 0 is $V^\perp$

$\{\vec{v_4}, \vec{v_5}\}$

$\mathbb{R}^n \xrightarrow{P} V$, $\vec{x}$

$P\vec{x} = \vec{x}$, eigenvalue 1 w/ eigenvector $\vec{x} \in V$

λ_1: ⇒ Espace for 1 is V

$P\vec{y} = \vec{0} = 0\vec{y}$

Eigenvalue is 0 w/ eigenvector $\vec{y} \in V^\perp$

$\mathbb{R}^5 = V \oplus V^\perp$

$\vec{v_1}, \vec{v_2}, \vec{v_3}, \vec{v_4}, \vec{v_5}$

$P = SDS^T$

$$S = \begin{bmatrix} | & | & | & | & | \\ \vec{v_1} & \vec{v_2} & \vec{v_3} & \vec{v_4} & \vec{v_5} \\ | & | & | & | & | \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & & & & \\ & 1 & & 0 & \\ & & 1 & & \\ & 0 & & 0 & \\ & & & & 0 \end{bmatrix}$$

λ_1: 1 almu(1) = 3

Basis E_1 $\{\vec{v_1}, \vec{v_2}, \vec{v_3}\}$ gemu(1) = 3

λ_2: 0 almu(0) = 2

Basis E_0 $\{\vec{v_4}, \vec{v_5}\}$ gemu(0) = 2